

```
In [1]: !git clone https://github.com/NTHung2034/PP-Final_Project.git
        %cd PP-Final_Project
```

```
Cloning into 'PP-Final_Project'...
remote: Enumerating objects: 1744, done.
remote: Counting objects: 100% (431/431), done.
remote: Compressing objects: 100% (366/366), done.
remote: Total 1744 (delta 147), reused 304 (delta 58), pack-reused 1313 (from 2)
Receiving objects: 100% (1744/1744), 154.15 MiB | 15.80 MiB/s, done.
Resolving deltas: 100% (840/840), done.
/content/PP-Final_Project
```

```
In [2]: !bash ./scripts/download_cifar10.sh
```

```
Downloading CIFAR-10 dataset...
Data directory: ./data
--2025-12-25 17:35:17-- https://www.cs.toronto.edu/~kriz/cifar-10-binary.tar.gz
Resolving www.cs.toronto.edu (www.cs.toronto.edu)... 128.100.3.30
Connecting to www.cs.toronto.edu (www.cs.toronto.edu)|128.100.3.30|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 170052171 (162M) [application/x-gzip]
Saving to: 'cifar-10-binary.tar.gz'

cifar-10-binary.tar 100%[=====>] 162.17M  8.74MB/s   in 20s

2025-12-25 17:35:37 (8.28 MB/s) - 'cifar-10-binary.tar.gz' saved [170052171/170052171]

CIFAR-10 dataset downloaded and extracted to ./data
Files:
-rw-r--r-- 1 2156 1103 30730000 Jun  4  2009 data_batch_1.bin
-rw-r--r-- 1 2156 1103 30730000 Jun  4  2009 data_batch_2.bin
-rw-r--r-- 1 2156 1103 30730000 Jun  4  2009 data_batch_3.bin
-rw-r--r-- 1 2156 1103 30730000 Jun  4  2009 data_batch_4.bin
-rw-r--r-- 1 2156 1103 30730000 Jun  4  2009 data_batch_5.bin
-rw-r--r-- 1 2156 1103 30730000 Jun  4  2009 test_batch.bin
```

```
In [3]: %cd external/thundersvm
        !mkdir build && cd build && cmake .. && make -j
        %cd ../../
```

```
/content/PP-Final_Project/external/thundersvm
CMake Deprecation Warning at CMakeLists.txt:1 (cmake_minimum_required):
  Compatibility with CMake < 3.10 will be removed from a future version of
  CMake.
```

Update the VERSION argument <min> value. Or, use the <min>...<max> syntax to tell CMake that the project requires at least <min> but has been updated to work with policies introduced by <max> or earlier.

```
-- The C compiler identification is GNU 11.4.0
-- The CXX compiler identification is GNU 11.4.0
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: /usr/bin/cc - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: /usr/bin/c++ - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Performing Test CMAKE_HAVE_LIBC_PTHREAD
-- Performing Test CMAKE_HAVE_LIBC_PTHREAD - Success
-- Found Threads: TRUE
-- Found OpenMP_C: -fopenmp (found version "4.5")
-- Found OpenMP_CXX: -fopenmp (found version "4.5")
-- Found OpenMP: TRUE (found version "4.5")
```

Compile with CUDA

```
CMake Warning (dev) at CMakeLists.txt:33 (find_package):
  Policy CMP0146 is not set: The FindCUDA module is removed. Run "cmake
  --help-policy CMP0146" for policy details. Use the cmake_policy command to
  set the policy and suppress this warning.
```

This warning is for project developers. Use -Wno-dev to suppress it.

```
-- Configuring done (1.4s)
-- Generating done (0.0s)
-- Build files have been written to: /content/PP-Final_Project/external/thundersvm/build
[ 3%] Building NVCC (Device) object src/thundersvm/CMakeFiles/thundersvm.dir/kernel/thundersvm_generated_smo_kernel.cu.o
[ 7%] Building NVCC (Device) object src/thundersvm/CMakeFiles/thundersvm.dir/kernel/thundersvm_generated_kernelmatrix_kernel.cu.o
[ 11%] Building CXX object src/thundersvm/CMakeFiles/thundersvm.dir/cmdparser.cpp.o
[ 15%] Building CXX object src/thundersvm/CMakeFiles/thundersvm.dir/model/nusvr.cpp.o
[ 19%] Building CXX object src/thundersvm/CMakeFiles/thundersvm.dir/kernelmatrix.cpp.o
[ 23%] Building CXX object src/thundersvm/CMakeFiles/thundersvm.dir/dataset.cpp.o
[ 26%] Building CXX object src/thundersvm/CMakeFiles/thundersvm.dir/model/nusvc.cpp.o
[ 30%] Building CXX object src/thundersvm/CMakeFiles/thundersvm.dir/model/oneclass_svc.cpp.o
[ 34%] Building CXX object src/thundersvm/CMakeFiles/thundersvm.dir/model/svmmodel.cpp.o
[ 38%] Building CXX object src/thundersvm/CMakeFiles/thundersvm.dir/model/svc.cpp.o
[ 42%] Building CXX object src/thundersvm/CMakeFiles/thundersvm.dir/model/svr.cpp.o
[ 46%] Building CXX object src/thundersvm/CMakeFiles/thundersvm.dir/solver/csmosolver.cpp.o
[ 50%] Building CXX object src/thundersvm/CMakeFiles/thundersvm.dir/solver/nusmosolver.cpp.o
[ 53%] Building CXX object src/thundersvm/CMakeFiles/thundersvm.dir/svm_R_interface.cpp.o
[ 57%] Building CXX object src/thundersvm/CMakeFiles/thundersvm.dir/syncarray.cpp.o
[ 61%] Building CXX object src/thundersvm/CMakeFiles/thundersvm.dir/syncmem.cpp.o
[ 65%] Building CXX object src/thundersvm/CMakeFiles/thundersvm.dir/svm_interface_api.cpp.o
[ 69%] Building CXX object src/thundersvm/CMakeFiles/thundersvm.dir/thundersvm-sikit.cpp.o
[ 73%] Building CXX object src/thundersvm/CMakeFiles/thundersvm.dir/util/common.cpp.o
[ 76%] Building CXX object src/thundersvm/CMakeFiles/thundersvm.dir/util/log.cpp.o
[ 80%] Building CXX object src/thundersvm/CMakeFiles/thundersvm.dir/util/metric.cpp.o
```

```

In file included from /content/PP-Final_Project/external/thundersvm/include/thundersvm/thunders
vm.h:13,
                from /content/PP-Final_Project/external/thundersvm/include/thundersvm/syncmem.
h:8,
                from /content/PP-Final_Project/external/thundersvm/src/thundersvm/syncmem.cpp:
5:
/content/PP-Final_Project/external/thundersvm/src/thundersvm/syncmem.cpp: In destructor 'thunde
r::SyncMem::~~SyncMem()':
/content/PP-Final_Project/external/thundersvm/include/thundersvm/util/common.h:37:44: warning:
'throw' will always call 'terminate' [8];https://gcc.gnu.org/onlinedocs/gcc/Warning-Options.h
tml#index-Wterminate-Wterminate]
    37 |         if(error == cudaErrorMemoryAllocation) throw std::bad_alloc(); \
        |                                             ^~~~~~
/content/PP-Final_Project/external/thundersvm/src/thundersvm/syncmem.cpp:29:17: note: in expans
ion of macro 'CUDA_CHECK'
    29 |         CUDA_CHECK(cudaFree(device_ptr));
        |         ^~~~~~
/content/PP-Final_Project/external/thundersvm/include/thundersvm/util/common.h:37:44: note: in
C++11 destructors default to 'noexcept'
    37 |         if(error == cudaErrorMemoryAllocation) throw std::bad_alloc(); \
        |                                             ^~~~~~
/content/PP-Final_Project/external/thundersvm/src/thundersvm/syncmem.cpp:29:17: note: in expans
ion of macro 'CUDA_CHECK'
    29 |         CUDA_CHECK(cudaFree(device_ptr));
        |         ^~~~~~
/content/PP-Final_Project/external/thundersvm/src/thundersvm/thundersvm-scikit.cpp: In function
'SvmModel* model_new(int)':
/content/PP-Final_Project/external/thundersvm/src/thundersvm/thundersvm-scikit.cpp:29:5: warnin
g: control reaches end of non-void function [8];https://gcc.gnu.org/onlinedocs/gcc/Warning-Op
tions.html#index-Wreturn-type-Wreturn-type]
    29 |     }
        |     ^
[ 84%] Linking CXX shared library ../../lib/libthundersvm.so
[ 84%] Built target thundersvm
[ 92%] Building CXX object src/thundersvm/CMakeFiles/thundersvm-predict.dir/thundersvm-predict.
cpp.o
[ 92%] Building CXX object src/thundersvm/CMakeFiles/thundersvm-train.dir/thundersvm-train.cpp.
o
[ 96%] Linking CXX executable ../../bin/thundersvm-predict
[100%] Linking CXX executable ../../bin/thundersvm-train
[100%] Built target thundersvm-predict
[100%] Built target thundersvm-train
/content/PP-Final_Project

```

```
In [4]: !cp CMakeThunder.txt CMakeLists.txt
```

```
In [5]: !mkdir -p build
        %cd build
        !cmake .. -DENABLE_CUDA=ON
        # compile
        !make
```

```

/content/PP-Final_Project/build
-- The CXX compiler identification is GNU 11.4.0
-- The C compiler identification is GNU 11.4.0
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: /usr/bin/c++ - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: /usr/bin/cc - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- Found CUDAToolkit: /usr/local/cuda/targets/x86_64-linux/include (found version "12.5.82")
-- Performing Test CMAKE_HAVE_LIBC_PTHREAD
-- Performing Test CMAKE_HAVE_LIBC_PTHREAD - Success
-- Found Threads: TRUE
-- CUDA Toolkit found - GPU support enabled
-- The CUDA compiler identification is NVIDIA 12.5.82 with host compiler GNU 11.4.0
-- Detecting CUDA compiler ABI info
-- Detecting CUDA compiler ABI info - done
-- Check for working CUDA compiler: /usr/local/cuda/bin/nvcc - skipped
-- Detecting CUDA compile features
-- Detecting CUDA compile features - done
-- GPU Naive targets: train_gpu_naive, extract_naive, thundersvm_naive
-- GPU Optimized v1 targets: train_gpu_opt_v1, extract_opt_v1
-- GPU Optimized v2 targets: train_gpu_opt_v2, extract_opt_v2
-- Configuring done (2.8s)
-- Generating done (0.1s)
-- Build files have been written to: /content/PP-Final_Project/build
[ 0%] Building CXX object CMakeFiles/train_cpu.dir/src/train_cpu.cpp.o
[ 1%] Building CXX object CMakeFiles/train_cpu.dir/src/data/cifar10_loader.cpp.o
[ 2%] Building CXX object CMakeFiles/train_cpu.dir/src/layers/conv2d_cpu.cpp.o
[ 3%] Building CXX object CMakeFiles/train_cpu.dir/src/layers/relu_cpu.cpp.o
[ 3%] Building CXX object CMakeFiles/train_cpu.dir/src/layers/maxpool_cpu.cpp.o
[ 4%] Building CXX object CMakeFiles/train_cpu.dir/src/layers/upsample_cpu.cpp.o
[ 5%] Building CXX object CMakeFiles/train_cpu.dir/src/models/autoencoder_cpu.cpp.o
[ 6%] Building CXX object CMakeFiles/train_cpu.dir/src/utils/logger.cpp.o
[ 7%] Building CXX object CMakeFiles/train_cpu.dir/src/utils/image_utils.cpp.o
[ 7%] Building CXX object CMakeFiles/train_cpu.dir/src/utils/memory_tracker.cpp.o
[ 8%] Linking CXX executable train_cpu
[ 8%] Built target train_cpu
[ 8%] Building CXX object CMakeFiles/extract_cpu.dir/src/models_svm/extract_cpu.cpp.o
[ 9%] Building CXX object CMakeFiles/extract_cpu.dir/src/data/cifar10_loader.cpp.o
[ 10%] Building CXX object CMakeFiles/extract_cpu.dir/src/layers/conv2d_cpu.cpp.o
[ 11%] Building CXX object CMakeFiles/extract_cpu.dir/src/layers/relu_cpu.cpp.o
[ 11%] Building CXX object CMakeFiles/extract_cpu.dir/src/layers/maxpool_cpu.cpp.o
[ 12%] Building CXX object CMakeFiles/extract_cpu.dir/src/layers/upsample_cpu.cpp.o
[ 13%] Building CXX object CMakeFiles/extract_cpu.dir/src/models/autoencoder_cpu.cpp.o
[ 14%] Linking CXX executable extract_cpu
[ 14%] Built target extract_cpu
[ 15%] Building CXX object CMakeFiles/train_svm_cpu.dir/src/models_svm/train_svm_cpu.cpp.o
[ 16%] Building CXX object CMakeFiles/train_svm_cpu.dir/src/data/cifar10_loader.cpp.o
[ 17%] Building CXX object CMakeFiles/train_svm_cpu.dir/external/svm.cpp.o
[ 18%] Linking CXX executable train_svm_cpu
[ 18%] Built target train_svm_cpu
[ 18%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/models_svm/extract_cpu.cpp.o
[ 19%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/data/cifar10_loader.cpp.o
[ 20%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/layers/conv2d_cpu.cpp.o
[ 21%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/layers/relu_cpu.cpp.o
[ 22%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/layers/maxpool_cpu.cpp.o
[ 22%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/layers/upsample_cpu.cpp.o

```

```
[ 23%] Building CXX object CMakeFiles/extract_features_cpu.dir/src/models/autoencoder_cpu.cpp.o
[ 24%] Linking CXX executable extract_features_cpu
[ 24%] Built target extract_features_cpu
[ 25%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/train_gpu_naive.cu.o
[ 26%] Building CXX object CMakeFiles/train_gpu_naive.dir/src/data/cifar10_loader.cpp.o
[ 26%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/layers_gpu_naive/conv2d_gpu_naive.cu.o
[ 27%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/layers_gpu_naive/maxpool_gpu_naive.cu.o
[ 28%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/layers_gpu_naive/relu_gpu_naive.cu.o
[ 29%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/layers_gpu_naive/upsample_gpu_naive.cu.o
[ 30%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/layers_gpu_naive/mse_loss_gpu_naive.cu.o
[ 30%] Building CUDA object CMakeFiles/train_gpu_naive.dir/src/models/autoencoder_gpu_naive.cu.o
[ 31%] Linking CUDA device code CMakeFiles/train_gpu_naive.dir/cmake_device_link.o
[ 32%] Linking CXX executable train_gpu_naive
[ 32%] Built target train_gpu_naive
[ 33%] Building CUDA object CMakeFiles/extract_naive.dir/src/models_svm/extract_naive.cu.o
[ 33%] Building CXX object CMakeFiles/extract_naive.dir/src/data/cifar10_loader.cpp.o
[ 34%] Building CUDA object CMakeFiles/extract_naive.dir/src/layers_gpu_naive/conv2d_gpu_naive.cu.o
[ 35%] Building CUDA object CMakeFiles/extract_naive.dir/src/layers_gpu_naive/maxpool_gpu_naive.cu.o
[ 36%] Building CUDA object CMakeFiles/extract_naive.dir/src/layers_gpu_naive/relu_gpu_naive.cu.o
[ 37%] Building CUDA object CMakeFiles/extract_naive.dir/src/layers_gpu_naive/upsample_gpu_naive.cu.o
[ 37%] Building CUDA object CMakeFiles/extract_naive.dir/src/layers_gpu_naive/mse_loss_gpu_naive.cu.o
[ 38%] Building CUDA object CMakeFiles/extract_naive.dir/src/models/autoencoder_gpu_naive.cu.o
[ 39%] Linking CUDA device code CMakeFiles/extract_naive.dir/cmake_device_link.o
[ 40%] Linking CXX executable extract_naive
[ 40%] Built target extract_naive
[ 41%] Building CUDA object CMakeFiles/thundersvm_naive.dir/src/models_svm/thundersvm_naive.cu.o
[ 42%] Building CXX object CMakeFiles/thundersvm_naive.dir/src/data/cifar10_loader.cpp.o
[ 42%] Building CUDA object CMakeFiles/thundersvm_naive.dir/src/layers_gpu_naive/conv2d_gpu_naive.cu.o
[ 43%] Building CUDA object CMakeFiles/thundersvm_naive.dir/src/layers_gpu_naive/maxpool_gpu_naive.cu.o
[ 44%] Building CUDA object CMakeFiles/thundersvm_naive.dir/src/layers_gpu_naive/relu_gpu_naive.cu.o
[ 45%] Building CUDA object CMakeFiles/thundersvm_naive.dir/src/layers_gpu_naive/upsample_gpu_naive.cu.o
[ 46%] Building CUDA object CMakeFiles/thundersvm_naive.dir/src/layers_gpu_naive/mse_loss_gpu_naive.cu.o
[ 46%] Building CUDA object CMakeFiles/thundersvm_naive.dir/src/models/autoencoder_gpu_naive.cu.o
[ 47%] Building CUDA object CMakeFiles/thundersvm_naive.dir/external/thunder_svm.cu.o
[ 48%] Linking CUDA device code CMakeFiles/thundersvm_naive.dir/cmake_device_link.o
[ 49%] Linking CXX executable thundersvm_naive
[ 49%] Built target thundersvm_naive
[ 50%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/train_gpu_opt_v1.cu.o
[ 50%] Building CXX object CMakeFiles/train_gpu_opt_v1.dir/src/data/cifar10_loader.cpp.o
[ 51%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/layers_gpu_opt_v1/conv2d_gpu_opt_v1.cu.o
[ 52%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/layers_gpu_opt_v1/relu_gpu_opt_v1.cu.o
[ 53%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/layers_gpu_opt_v1/maxpool_gpu_opt_v1.cu.o
```

```
pt_v1.cu.o
[ 54%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/layers_gpu_opt_v1/upsample_gpu_opt_v1.cu.o
[ 54%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/layers_gpu_opt_v1/mse_loss_gpu_opt_v1.cu.o
[ 55%] Building CUDA object CMakeFiles/train_gpu_opt_v1.dir/src/models/autoencoder_gpu_opt_v1.cu.o
[ 56%] Linking CUDA device code CMakeFiles/train_gpu_opt_v1.dir/cmake_device_link.o
[ 57%] Linking CXX executable train_gpu_opt_v1
[ 57%] Built target train_gpu_opt_v1
[ 57%] Building CUDA object CMakeFiles/extract_opt_v1.dir/src/models_svm/extract_opt_v1.cu.o
[ 58%] Building CXX object CMakeFiles/extract_opt_v1.dir/src/data/cifar10_loader.cpp.o
[ 59%] Building CUDA object CMakeFiles/extract_opt_v1.dir/src/layers_gpu_opt_v1/conv2d_gpu_opt_v1.cu.o
[ 60%] Building CUDA object CMakeFiles/extract_opt_v1.dir/src/layers_gpu_opt_v1/relu_gpu_opt_v1.cu.o
[ 61%] Building CUDA object CMakeFiles/extract_opt_v1.dir/src/layers_gpu_opt_v1/maxpool_gpu_opt_v1.cu.o
[ 61%] Building CUDA object CMakeFiles/extract_opt_v1.dir/src/layers_gpu_opt_v1/upsample_gpu_opt_v1.cu.o
[ 62%] Building CUDA object CMakeFiles/extract_opt_v1.dir/src/layers_gpu_opt_v1/mse_loss_gpu_opt_v1.cu.o
[ 63%] Building CUDA object CMakeFiles/extract_opt_v1.dir/src/models/autoencoder_gpu_opt_v1.cu.o
[ 64%] Linking CUDA device code CMakeFiles/extract_opt_v1.dir/cmake_device_link.o
[ 64%] Linking CXX executable extract_opt_v1
[ 64%] Built target extract_opt_v1
[ 64%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/train_gpu_opt_v2.cu.o
[ 65%] Building CXX object CMakeFiles/train_gpu_opt_v2.dir/src/data/cifar10_loader.cpp.o
[ 66%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/layers_gpu_opt_v2/conv2d_gpu_opt_v2.cu.o
[ 67%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/layers_gpu_opt_v2/relu_gpu_opt_v2.cu.o
[ 68%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/layers_gpu_opt_v2/maxpool_gpu_opt_v2.cu.o
[ 68%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/layers_gpu_opt_v2/upsample_gpu_opt_v2.cu.o
[ 69%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/layers_gpu_opt_v2/mse_loss_gpu_opt_v2.cu.o
[ 70%] Building CUDA object CMakeFiles/train_gpu_opt_v2.dir/src/models/autoencoder_gpu_opt_v2.cu.o
[ 71%] Linking CUDA device code CMakeFiles/train_gpu_opt_v2.dir/cmake_device_link.o
[ 71%] Linking CXX executable train_gpu_opt_v2
[ 71%] Built target train_gpu_opt_v2
[ 72%] Building CUDA object CMakeFiles/extract_opt_v2.dir/src/models_svm/extract_opt_v2.cu.o
[ 73%] Building CXX object CMakeFiles/extract_opt_v2.dir/src/data/cifar10_loader.cpp.o
[ 74%] Building CUDA object CMakeFiles/extract_opt_v2.dir/src/layers_gpu_opt_v2/conv2d_gpu_opt_v2.cu.o
[ 75%] Building CUDA object CMakeFiles/extract_opt_v2.dir/src/layers_gpu_opt_v2/relu_gpu_opt_v2.cu.o
[ 75%] Building CUDA object CMakeFiles/extract_opt_v2.dir/src/layers_gpu_opt_v2/maxpool_gpu_opt_v2.cu.o
[ 76%] Building CUDA object CMakeFiles/extract_opt_v2.dir/src/layers_gpu_opt_v2/upsample_gpu_opt_v2.cu.o
[ 77%] Building CUDA object CMakeFiles/extract_opt_v2.dir/src/layers_gpu_opt_v2/mse_loss_gpu_opt_v2.cu.o
[ 78%] Building CUDA object CMakeFiles/extract_opt_v2.dir/src/models/autoencoder_gpu_opt_v2.cu.o
[ 78%] Linking CUDA device code CMakeFiles/extract_opt_v2.dir/cmake_device_link.o
[ 79%] Linking CXX executable extract_opt_v2
[ 79%] Built target extract_opt_v2
[ 80%] Building CUDA object CMakeFiles/test_comparison_cpu_gpu.dir/tests/comparison_cpu_gpu.cu.o
```

```

o
[ 81%] Building CXX object CMakeFiles/test_comparison_cpu_gpu.dir/src/data/cifar10_loader.cpp.o
[ 81%] Building CXX object CMakeFiles/test_comparison_cpu_gpu.dir/src/layers/conv2d_cpu.cpp.o
[ 82%] Building CXX object CMakeFiles/test_comparison_cpu_gpu.dir/src/layers/relu_cpu.cpp.o
[ 83%] Building CXX object CMakeFiles/test_comparison_cpu_gpu.dir/src/layers/maxpool_cpu.cpp.o
[ 84%] Building CXX object CMakeFiles/test_comparison_cpu_gpu.dir/src/layers/upsample_cpu.cpp.o
[ 85%] Building CXX object CMakeFiles/test_comparison_cpu_gpu.dir/src/models/autoencoder_cpu.cpp.o
[ 85%] Building CUDA object CMakeFiles/test_comparison_cpu_gpu.dir/src/layers_gpu_naive/conv2d_gpu_naive.cu.o
[ 86%] Building CUDA object CMakeFiles/test_comparison_cpu_gpu.dir/src/layers_gpu_naive/maxpool_gpu_naive.cu.o
[ 87%] Building CUDA object CMakeFiles/test_comparison_cpu_gpu.dir/src/layers_gpu_naive/relu_gpu_naive.cu.o
[ 88%] Building CUDA object CMakeFiles/test_comparison_cpu_gpu.dir/src/layers_gpu_naive/upsample_gpu_naive.cu.o
[ 88%] Building CUDA object CMakeFiles/test_comparison_cpu_gpu.dir/src/layers_gpu_naive/mse_loss_gpu_naive.cu.o
[ 89%] Building CUDA object CMakeFiles/test_comparison_cpu_gpu.dir/src/models/autoencoder_gpu_naive.cu.o
[ 90%] Building CUDA object CMakeFiles/test_comparison_cpu_gpu.dir/src/layers_gpu_opt_v1/conv2d_gpu_opt_v1.cu.o
[ 91%] Building CUDA object CMakeFiles/test_comparison_cpu_gpu.dir/src/layers_gpu_opt_v1/relu_gpu_opt_v1.cu.o
[ 92%] Building CUDA object CMakeFiles/test_comparison_cpu_gpu.dir/src/layers_gpu_opt_v1/maxpool_gpu_opt_v1.cu.o
[ 92%] Building CUDA object CMakeFiles/test_comparison_cpu_gpu.dir/src/layers_gpu_opt_v1/upsample_gpu_opt_v1.cu.o
[ 93%] Building CUDA object CMakeFiles/test_comparison_cpu_gpu.dir/src/layers_gpu_opt_v1/mse_loss_gpu_opt_v1.cu.o
[ 94%] Building CUDA object CMakeFiles/test_comparison_cpu_gpu.dir/src/models/autoencoder_gpu_opt_v1.cu.o
[ 95%] Building CUDA object CMakeFiles/test_comparison_cpu_gpu.dir/src/layers_gpu_opt_v2/conv2d_gpu_opt_v2.cu.o
[ 95%] Building CUDA object CMakeFiles/test_comparison_cpu_gpu.dir/src/layers_gpu_opt_v2/relu_gpu_opt_v2.cu.o
[ 96%] Building CUDA object CMakeFiles/test_comparison_cpu_gpu.dir/src/layers_gpu_opt_v2/maxpool_gpu_opt_v2.cu.o
[ 97%] Building CUDA object CMakeFiles/test_comparison_cpu_gpu.dir/src/layers_gpu_opt_v2/upsample_gpu_opt_v2.cu.o
[ 98%] Building CUDA object CMakeFiles/test_comparison_cpu_gpu.dir/src/layers_gpu_opt_v2/mse_loss_gpu_opt_v2.cu.o
[ 99%] Building CUDA object CMakeFiles/test_comparison_cpu_gpu.dir/src/models/autoencoder_gpu_opt_v2.cu.o
[ 99%] Linking CUDA device code CMakeFiles/test_comparison_cpu_gpu.dir/cmake_device_link.o
[100%] Linking CXX executable test_comparison_cpu_gpu
[100%] Built target test_comparison_cpu_gpu

```

In [6]: `!./train_gpu_naive`

CIFAR-10 AUTOENCODER - NAIVE GPU IMPLEMENTATION (PHASE 2)

Using GPU: Tesla T4
Compute Capability: 7.5
Total Global Memory: 15095 MB

Loading CIFAR-10 training data...
Loaded data_batch_1.bin
Loaded data_batch_2.bin
Loaded data_batch_3.bin
Loaded data_batch_4.bin
Loaded data_batch_5.bin
Training data loaded: 50000 images

=====

NAIVE GPU AUTOENCODER TRAINING

=====

Configuration:

Batch size: 64
Epochs: 1
Learning rate: 0.001
Training samples: 50000
Save directory: /content/PP-Final_Project/models/saved_weights_gpu_naive

=====

Batches per epoch: 781
VRAM usage: 308 MB

Epoch [1/1] | Loss: 0.078121 | Time: 450.78s | Avg: 450.78s/epoch

=====

TRAINING COMPLETED

=====

Total training time: 450.78s
Average time per epoch: 450.78s
Initial loss: 0.078121
Final loss: 0.078121
Loss reduction: 0.000000%

=====

Final VRAM usage: 308 MB

Saving model weights to: /content/PP-Final_Project/models/saved_weights_gpu_naive

Saved conv1 weights: 6912 params, bias: 256 params
Saved conv2 weights: 294912 params, bias: 128 params
Saved conv3 weights: 147456 params, bias: 128 params
Saved conv4 weights: 294912 params, bias: 256 params
Saved conv5 weights: 6912 params, bias: 3 params

Model weights saved successfully!

Training summary saved to: /content/PP-Final_Project/models/saved_weights_gpu_naive/training_summary.txt

TRAINING COMPLETED!

In [7]: !./thundersvm_naive

CIFAR-10 AUTOENCODER - NAIVE GPU + SVM (PHASE 4)

=== Step 1: Loading Dataset ===

Loading CIFAR-10 training data...

Loaded data_batch_1.bin

Loaded data_batch_2.bin

Loaded data_batch_3.bin

Loaded data_batch_4.bin

Loaded data_batch_5.bin

Training data loaded: 50000 images

Loading CIFAR-10 test data...

Test data loaded: 10000 images

Training images: 50000

Test images: 10000

=== Step 2: Loading Encoder Weights ===

Encoder weights loaded

=== Step 3: Extracting Features ===

Train batch [50/500]

Train batch [100/500]

Train batch [150/500]

Train batch [200/500]

Train batch [250/500]

Train batch [300/500]

Train batch [350/500]

Train batch [400/500]

Train batch [450/500]

Train batch [500/500]

Test batch [20/100]

Test batch [40/100]

Test batch [60/100]

Test batch [80/100]

Test batch [100/100]

Feature extraction completed: 36.57s

Normalizing features...

=== Step 4: Training SVM ===

Training SVM classifier with ThunderSVM (GPU)...

Samples: 50000, Features: 8192

Preparing training data...

Written 50000/50000 samples

Data preparation completed in 194.83 seconds

=====

Starting ThunderSVM GPU training...

=====

2025-12-25 17:50:44,470 INFO [default] loading dataset from file "/tmp/thundersvm_train.txt"

2025-12-25 17:53:49,407 INFO [default] #instances = 50000, #features = 8192

2025-12-25 17:53:51,355 INFO [default] #classes = 10

2025-12-25 17:53:55,556 INFO [default] total memory size is 0.910573 max mem size is 8

2025-12-25 17:53:55,557 INFO [default] free mem is 7.08943

2025-12-25 17:53:55,562 INFO [default] working set size = 1024

2025-12-25 17:53:55,563 INFO [default] training start

Error: radix_sort: failed on 1st step: cudaErrorUnsupportedPtxVersion: the provided PTX was compiled with an unsupported toolchain.